

Ashbourne & Co. Business Intelligence Brief

1. Executive Overview & Company History

Founded in 2011 by Alexander Ashbourne (Current Chairman), **Ashbourne & Co.** began as a single boutique in London's Covent Garden. Over the last decade, the brand has organically grown into a leading premium Western clothing retailer in the UK. By fusing rugged heritage styling with sharp British tailoring, the company now serves Men, Women, and Kids through a footprint of 15+ physical locations—including high-profile Flagships, regional Boutiques, and premium Outlets—alongside a rapidly scaling E-commerce platform.

The Current Business Challenge:

Under the current leadership of CEO Eleanor Vance and Managing Director Julian Wright, Ashbourne & Co. has experienced impressive top-line revenue growth. However, the executive board has noted a concerning compression in profit margins. Post-pandemic shifts in consumer behaviour have led to an imbalance in inventory allocation between digital and physical channels. Furthermore, the company relies heavily on promotional discounting in Outlet stores and aggressive digital marketing campaigns, but the true ROI and cannibalization effects of these strategies remain opaque.

Strategic Objectives of this Analysis:

As the board prepares for a major European expansion in 2026, they require a rigorous, data-driven audit of omni-channel performance spanning January 2023 through December 2025. The insights derived from this project will directly influence next year's operational budgets, store layout planning, and digital marketing spend.

Technical Approach (SQL vs. Python):

The Data Engineering team mandates that core financial metric extraction, transactional reporting, and baseline aggregations be performed in **SQL** to ensure they run natively and efficiently on our enterprise data warehouse. Conversely, **Python** must be leveraged for multi-variate Exploratory Data Analysis (EDA), advanced customer segmentations (such as RFM analysis), and statistical hypothesis testing, as these tasks require statistical libraries (e.g., pandas, scipy) that surpass the capabilities of standard SQL.

2. Tableau Dashboard Specification

The BI team must construct a 3-page interactive Tableau dashboard utilizing the central Order's fact table and all related dimension tables. Each page requires a specific suite of visualizations to answer distinct operational questions.

Page 1: Executive Sales & Margin Overview

- **Primary Audience:** C-Suite Executives (CEO, Chairman).
- **Filters/Slicers:** Date Range (Year/Quarter), Store Type, Category.
- **Visual 1 (Top Banner KPIs):** Total Net Sales (£), Gross Margin (%), Total Orders, Average Order Value (AOV), Return Rate (%). Include YoY % change indicators.
- **Visual 2 (Dual-Axis Line Chart):** *Monthly Revenue & Profit Trends*. Identify macroeconomic seasonality and periods of margin erosion.
- **Visual 3 (Waterfall Chart):** *Cumulative Monthly Profit*. Track net profit accumulation across the year, highlighting the exact financial impact of high-performing and loss-making months.
- **Visual 4 (Geospatial Map):** *Sales by Geography*. UK map plotting Store City to Net Sales.
- **Visual 5 (Stacked Bar Chart):** *Revenue by Category & Season*. Track the contribution of 'Autumn/Winter' collections vs. 'Core' lines over time.

Page 2: Customer Loyalty & Retention Analytics

- **Primary Audience:** Marketing & CRM Directors.
- **Filters/Slicers:** Customer Loyalty Tier, Income Bracket, Gender.
- **Visual 1 (KPI Cards):** Total Active Customers, Repeat Purchase Rate, Average Customer Lifetime Value (CLV).
- **Visual 2 (Scatter Plot):** *Customer Age vs. Total Spend*. Detail by Customer ID, colored by Loyalty Tier, to find demographic sweet spots.
- **Visual 3 (Donut Chart):** *Revenue by Acquisition Source*. Determine which marketing channels initially bring in the highest-value clients.
- **Visual 4 (Bar Chart):** *Average CLV by Loyalty Tier*. Prove the financial value of moving a customer from Silver to Platinum.
- **Visual 5 (Density Heatmap):** *Purchasing Patterns*. Heatmap showing volume of orders by Month and Day of Month to optimize CRM email timings.
- **Visual 6 (Dynamic Top-N Table):** *Top N Customers*. Interactive table (N=10/20/50) showing Name, Tier, Total Spend, and preferred Store.

Page 3: Store Operations & Marketing Efficiency

- **Primary Audience:** Operations & Merchandising Managers.
- **Filters/Slicers:** Store Name, Campaign Channel, Material.
- **Visual 1 (Horizontal Bar):** *Store Performance (Gross vs. Net)*. Sorted by Net Sales. Tooltip must display Square Footage for spatial efficiency context.
- **Visual 2 (Tree Map):** *Product Sub-category Profitability*. Size = Sales Volume, Color = Profit Margin. Identify margin winners like 'Cashmere' vs. 'Denim'.
- **Visual 3 (Bullet Chart):** *Marketing Campaign ROI*. Compare specific Campaign Budget (target line) to attributed Gross Sales (actual bar).
- **Visual 4 (Line Chart):** *Return Rate Volatility by Month*. Highlight spikes in returns following major promotional events.
- **Visual 5 (Scatter Plot):** *Inventory Level vs. Sales Velocity*. Compare current physical stock levels against 90-day sales volume to flag overstock risks.

3. Statistical Hypothesis Testing

The executive board has formulated several operational assumptions. Use statistical testing in Python to validate these claims and provide actionable business interpretations.

1. Return Rate Crisis (Significance Level: $\alpha = 0.01$)

Business Problem: Operations suspects our lenient E-Commerce return policy is severely impacting margins compared to physical stores, potentially requiring a fee for online returns.

Expectation: Determine if the proportion of returns online is statistically greater than in-store.

2. Gender-Based Ad Targeting (Significance Level: $\alpha = 0.05$)

Business Problem: Marketing wants to allocate a £500k ad budget but is unsure whether to heavily skew it toward Male or Female collections based on basket sizes.

Expectation: Test if there is a statistically significant difference in Average Order Value (AOV) between Male and Female shoppers.

3. Inventory Seasonality (Significance Level: $\alpha = 0.025$)

Business Problem: Merchandising believes Kids' clothing sales are less sensitive to seasonal shifts (Autumn/Winter vs Spring/Summer) than Adult categories.

Expectation: Test for independence between Product Category and Purchasing Season to guide next year's manufacturing orders.

4. Format Margin Discrepancy (Significance Level: $\alpha = 0.10$)

Business Problem: The CFO wants to pause new Outlet openings, arguing that heavy discounting in Outlets and Boutiques makes them fundamentally less profitable than Flagships.

Expectation: Analyze if average profit margins significantly differ across the three physical store formats.

5. Discount Cannibalization (Significance Level: $\alpha = 0.01$)

Business Problem: We frequently offer 20-40% discounts hoping customers buy more items per transaction. Sales directors fear we are just losing money on single-item orders.

Expectation: Evaluate the linear correlation between discount percentage and the quantity of items purchased per order.

6. Quality Expectations (Significance Level: $\alpha = 0.05$)

Business Problem: Customer Service reports that our Ultra-High income clientele are leaving harsher reviews on premium items.

Expectation: Check for a monotonic relationship between ordinal income brackets and the 1-5 star ratings left on orders.

7. Pricing & Cost Multicollinearity (Threshold: $|r| > 0.75$)

Business Problem: Before building predictive pricing models, the data science team needs to understand the intrinsic relationships between our financial metrics.

Expectation: Generate a correlation matrix to flag multicollinearity risks among unit_cost, retail_price, quantity, discount_pct, and profit.

4. Python Exploratory Data Analysis (EDA)

Perform these 20 EDA tasks to uncover hidden trends before modeling.

Sales & Profitability

1. Plot the monthly distribution of Net Sales (line chart) to verify Q4 seasonal reliance.
2. Create a boxplot of Profit per order by Category to spot extreme margin outliers.
3. Calculate and visualize the overall Return Rate percentage (pie chart).
4. Plot a histogram of Discount Percentages applied across all transactions.
5. Compare Gross vs. Net Sales for the top 5 performing sub-categories (grouped bar).

Customer Behavior

6. Calculate RFM (Recency, Frequency, Monetary) scores and plot the segment sizes.
7. Plot a histogram of Customer Age (with KDE) to understand the core demographic.
8. Use a count plot to show the distribution of customers across Loyalty Tiers.
9. Analyze Average Order Value by Income Bracket (bar chart with error bars).
10. Visualize order counts by Acquisition Source (horizontal bar chart).

Stores & Operations

1. Calculate average Delivery Time (days) for E-commerce and plot its distribution.
2. Compare total physical Stock Level across different Store Types (donut chart).
3. Scatter plot Store Square Footage against Total Revenue for spatial efficiency.
4. Plot a stacked bar chart of Order Status (Completed/Returned) by Store City.
5. Identify the top 3 stores with the highest volume of low-rating (1-2) orders.

Products & Marketing

16. Visualize the distribution of Unit Costs vs Retail Prices (joint plot).
17. Plot total revenue generated by each Marketing Campaign channel (bar chart).
18. Analyze which Material type yields the highest average Rating (bar plot).
19. Compare Campaign Budget vs Actual Revenue (scatter plot + regression line).
20. Create a time-series plot marking Campaign Start Dates against daily volume.

5. SQL Data Engineering & Reporting Tasks

The following queries must be written for execution in Snowflake to build out our automated reporting tables.

Sales & Profitability Analytics

1. **Category Revenue:** Calculate total `net_sales` and quantity sold for each product category, sorted descending by revenue.
2. **Discount Impact:** Find the average `discount_pct` applied specifically to orders processed at 'Outlet' store types.
3. **Store Profitability (Joins):** Join `Orders` and `Stores` to calculate total profit generated by each `store_name`. Exclude 'E-Commerce'.
4. **Seasonal Performance:** Extract the Month from `order_date` and aggregate total `gross_sales` for 'Autumn/Winter' vs. 'Spring/Summer' products per month.
5. **Month-Over-Month Growth (CTEs & Window Functions):** Create a CTE for total monthly revenue. Use the `LAG()` function to calculate MoM % growth rate for the 2024 calendar year.
6. **Running Cumulative Profit:** Calculate the cumulative running total of `profit` over the year 2025, ordered chronologically by `order_date`.

Customer Behavior & Retention

7. **High-Value VIPs:** Retrieve the names and emails of all Platinum tier customers located in the flagship city of 'London'.
8. **Customer LTV:** Group by customer_id to calculate Total Spend and Total Order Count. Use HAVING to filter for customers spending over £1,500.
9. **Top Customer per City (Window Functions):** Use the RANK() function to identify the single highest-spending customer in each city.
10. **Customer Churn Risk:** Write a query to find customers who placed an order in 2023 or 2024 but have not placed any orders in the last 12 months (relative to Dec 31, 2025).
11. **Market Basket Indicator:** Using self-joins, identify the top 5 most frequent sub_category combinations purchased by the same customer on the same date.

Stores & Operations

12. **Low Stock Alerts:** List all products currently showing a stock_level below 20 units across any physical store location.
13. **Return Volume:** Count the total number of 'Returned' orders and calculate this as a percentage of overall transaction volume.
14. **Delivery SLA:** For all E-commerce orders, calculate the date difference between order and delivery dates. Find the average delivery time per month.
15. **Unsold Inventory:** Use a LEFT JOIN to find any items in the Products table that have never been recorded in the Orders fact table.

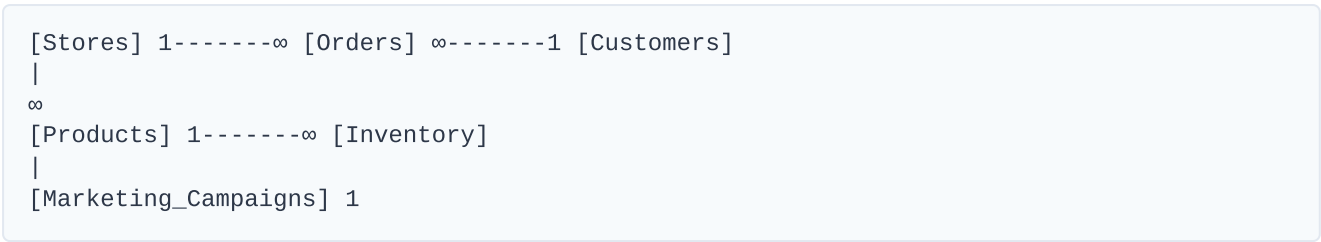
Products & Marketing

16. **Major Campaigns:** Select all marketing campaigns that launched in 2024 with an allocated budget exceeding £20,000.
17. **Campaign ROI:** Join Campaigns to Orders. Calculate net revenue attributed to each campaign, subtract the budget, and output the true monetary ROI.
18. **Material Margins:** Find the top 3 product material types by average profit margin, ensuring only materials with at least 100 historical orders are included.
19. **Pareto Principle (80/20 Rule):** Use a cumulative window function to determine if the top 20% of highest-selling products account for 80% of total revenue.
20. **Pricing Bands:** Categorize products into 'Entry', 'Mid', and 'Luxury' based on retail_price quartiles, and count total units sold in each band.

Appendix A: Master Data Dictionary & Schema

The following structures define the Ashbourne & Co. data warehouse star schema.

A1. Entity Relationship Overview



A2. Customers Table

Column	Type	Key	Description
customer_id	INT	PK	Unique identifier for the customer.
gender	VARCHAR		Self-reported gender identity.
age	INT		Customer age in years.
income_bracket	VARCHAR		Estimated bracket (Low, Medium, High, Ultra-High).
acquisition_source	VARCHAR		Channel through which customer was acquired.
city	VARCHAR		Customer's primary shipping city.
join_date	DATE		Date of first profile registration.
loyalty_tier	VARCHAR		System-assigned tier (Silver, Gold, Platinum).

A3. Products Table

Column	Type	Key	Description
product_id	INT	PK	Unique SKU identifier.
product_name	VARCHAR		Standardized product display name.
category	VARCHAR		Top-level category (Men, Women, Kids).
sub_category	VARCHAR		Secondary category (Outerwear, Denim, etc.).
season	VARCHAR		Release season (Spring/Summer, Autumn/Winter).
material	VARCHAR		Primary garment material composition.
unit_cost	DECIMAL		Manufacturing/landed cost in GBP (£).
retail_price	DECIMAL		Standard retail selling price in GBP (£).

A4. Stores & Operations Tables

Table & Column	Type	Key	Description
Stores Table			
store_id	INT	PK	Unique branch identifier.
store_type	VARCHAR		Flagship, Boutique, Outlet, E-Commerce.
square_footage	INT		Physical size (0 for online).
Marketing_Campaigns Table			
campaign_id	INT	PK	Unique campaign identifier.
budget	DECIMAL		Allocated ad spend in GBP (£).
Inventory Table			
inventory_id	INT	PK	Unique record identifier.
stock_level	INT		Current physical units on hand.

A5. Orders (Fact Table)

Column	Type	Key	Description
order_id	INT	PK	Unique transaction identifier.
customer_id	INT	FK	References Customers.
product_id	INT	FK	References Products.
store_id	INT	FK	References Stores.
campaign_id	INT	FK	References Campaigns (Nullable).
order_date	DATE		Date transaction was processed.
quantity	INT		Units purchased in this line item.
discount_pct	DECIMAL		Percentage discount applied (0.00 to 1.00).
gross_sales	DECIMAL		Quantity * Retail Price.
net_sales	DECIMAL		Gross Sales - Discount Amount.
profit	DECIMAL		Net Sales - COGS.
return_status	VARCHAR		'Completed' or 'Returned'.
rating	INT		Customer feedback score (1-5).